

```

from google.colab import files
uploaded = files.upload()

<IPython.core.display.HTML object>

Saving wav2vec2_finetuned.zip to wav2vec2_finetuned.zip

!unzip wav2vec2_finetuned.zip
Archive: wav2vec2_finetuned.zip
  creating: wav2vec2_finetuned/
  inflating: wav2vec2_finetuned/model.safetensors
  inflating: wav2vec2_finetuned/config.json
  inflating: wav2vec2_finetuned/processor_config.json
  inflating: wav2vec2_finetuned/tokenizer_config.json
  inflating: wav2vec2_finetuned/vocab.json
  inflating: wav2vec2_finetuned/training_args.bin

model_path = "/content/wav2vec2_finetuned"

from google.colab import files
uploaded = files.upload()
audio_path = list(uploaded.keys())[0]

<IPython.core.display.HTML object>

Saving sample_audio.wav to sample_audio.wav

from IPython.display import Audio, display
# !ffmpeg -i "$audio_path" -ar 16000 -ac 1 converted.wav -y

display(Audio(audio_path))

<IPython.lib.display.Audio object>

!pip install -q openai-whisper jiwer

_____ 0.0/803.2 kB ? eta -:--:--
_____ 92.2/803.2 kB 2.5 MB/s eta
0:00:01 _____ 266.2/803.2 kB 4.0
MB/s eta 0:00:01 _____ 665.6/803.2
kB 6.4 MB/s eta 0:00:01 _____
803.2/803.2 kB 6.6 MB/s eta 0:00:00
ents to build wheel ... etadata (pyproject.toml) ...
_____ 3.1/3.1 MB 46.7 MB/s eta
0:00:00
_____ 197.7/197.7 MB 6.3 MB/s eta
0:00:00
l) ...

import whisper
from jiwer import wer

```

```

whisper_model = whisper.load_model("base") # or your saved/fine-tuned
model
result = whisper_model.transcribe(audio_path)
print("Whisper Transcript:")
pred_text = result["text"].upper()

print(pred_text)

/usr/local/lib/python3.12/dist-packages/whisper/transcribe.py:132:
UserWarning: FP16 is not supported on CPU; using FP32 instead
  warnings.warn("FP16 is not supported on CPU; using FP32 instead")

Whisper Transcript:
MR. QUILTER IS THE APOSTLE OF THE MIDDLE CLASSES, AND WE ARE GLAD TO
WELCOME HIS GOSPEL.

from transformers import pipeline

asr_pipe = pipeline(
    "automatic-speech-recognition",
    model=model_path
)

wav2vec_result = asr_pipe(audio_path)

print("Wav2Vec2/Transformer Transcript:")
print(wav2vec_result["text"])

{"model_id": "d84f9e2b643b452fb83ee225ef5ccf46", "version_major": 2, "vers
ion_minor": 0}

Wav2Vec2/Transformer Transcript:
MISTER QUILTER IS THE APOSTLE OF THE MIDDLE CLASSES AND WE ARE GLAD TO
WELCOME HIS GOSPEL

print("==== WHISPER ====")
print(pred_text)

print("\n==== WAV2VEC2 / TRANSFORMER ====")
wav2vec_pred = wav2vec_result["text"]
print(wav2vec_pred)

==== WHISPER ====
MR. QUILTER IS THE APOSTLE OF THE MIDDLE CLASSES, AND WE ARE GLAD TO
WELCOME HIS GOSPEL.

==== WAV2VEC2 / TRANSFORMER ====
MISTER QUILTER IS THE APOSTLE OF THE MIDDLE CLASSES AND WE ARE GLAD TO
WELCOME HIS GOSPEL

```

to calculate the wer of this please input the actual script here

```
actual_text = "MISTER QUILTER IS THE APOSTLE OF THE MIDDLE CLASSES AND  
WE ARE GLAD TO WELCOME HIS GOSPEL"
```

```
whisper_wer = wer(actual_text, pred_text)  
wav2vec_wer = wer(actual_text, wav2vec_pred)
```

```
print("Whisper WER %:", whisper_wer * 100)  
print("Wav2Vec2 WER %:", wav2vec_wer * 100)
```

```
Whisper WER %: 17.647058823529413  
Wav2Vec2 WER %: 0.0
```